Chapter 12: The Collapse of the Phillips Curve and the Inflation Bias

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1 Overview: The Collapse of the Phillips curve

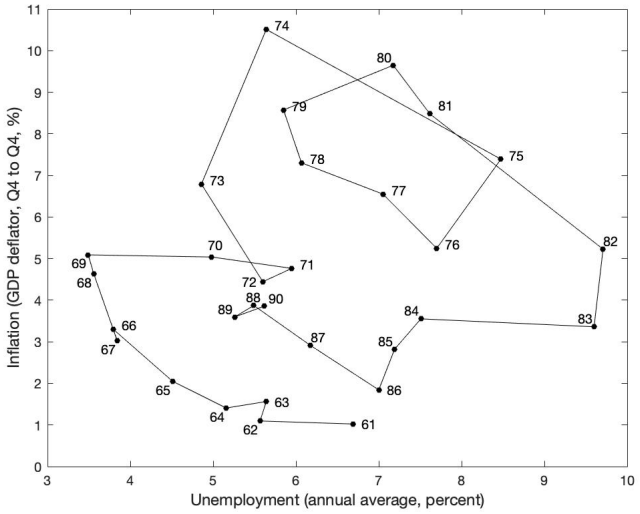
The Phillips curve was believed to be be stable in the 1960s. You can see a stable Phillips curve traced out during the years 1961-69 in Figure 1. During the 1970s, however, this relationship broke down, as Figure 1 shows. That this relationship would break down as soon as the Federal Reserve tried to exploit it was *predicted* by Milton Friedman and Edmund Phelps in the 1960s. They suggested that, rather than taking $\pi_t = \kappa(\hat{Y}_t - \hat{Y}_t^n)$ as a structural equation, a more realistic model should take into account that firms, in setting their prices, forecast future economic conditions. Hence, a more realistic Phillips curve would instead be described by an equation such as:

$$\pi_t = \kappa(\hat{Y}_t - \hat{Y}_t^n) + E_{t-1}\pi_t \tag{1}$$

The important prediction of this equation is that as soon as inflation is anticipated, i.e. $\pi_t = E_{t-1}\pi_t$, then $\hat{Y}_t = \hat{Y}_t^n$ and the Phillips curve relationship breaks down, just as Figure 1 suggests, so that there is no gain in output from inflation. According to this narrative the Great Inflation of the 1970s did not accomplish anything – except for, simply, generating greater inflation.

In this chapter we show that if we change the model from previous chapter so that the fraction of firms that indexed their price to the past-period price level instead set their price one period in advance, trying to forecast economic conditions, then equation (1) emerges. Equation (1) has been called the *New Classical Phillips Curve*.

But if inflation has no effect on output, once anticipated, why would a government ever choose to increase inflation in the first place? We conclude this chapter by showing a famous result from Kydland and Prescott, which was cited in their Nobel prize announcement. This fundamental result has been termed the inflation bias. As we will see, a government that sets policy under discretion,



i.e. a government unable to commit to future policies, delivers high inflation without any gains in output. According to this theory, then, the Great Inflation of the 1970s was due to policy makers starting to act in a *discretionary manner*, abandoning the more rule-based policy frameworks that guaranteed low inflation in previous decades.

1.1 The New Classical Phillips Curve

Consider the same model as before but with a different assumption: a fraction γ of firms set prices contemporaneously, p_t^{flex}, while the remaining firms $(1-\gamma)$ set prices one period in advance, p_t^{fix}, but do so by making a forecast about the next period's economic conditions. This assumption will imply a New Classical Phillips curve as in (1), as we will now see.

The firms that set their prices flexibly earn the following period profit:

$$Z_t^{flex}(i) = p_t(i)^{1-\theta_t} P_t^{\theta_t} Y_t - \frac{W_t p_t(i)^{-\theta_t} P_t^{\theta_t} Y_t}{A_t}$$

whose maximization yields the first-order condition:

$$(1-\theta_t) p_t(i)^{-\theta_t} P_t^{\theta_t} Y_t + \theta_t \frac{W_t P_t^{\theta} Y_t}{A_t} p_t(i)^{-\theta_t - 1} = 0$$

Those firms that set their price one period in advance, on the other hand, maximize the same profit function one period earlier, thus applying the stochastic

discount factor:

$$Z_t^{fix}(i) = E_{t-1}\beta\frac{C_t^{-\sigma}\xi_t}{P_t}\left\{p_t(i)^{1-\theta_t}P_t^{\theta_t}Y_t - \frac{W_t p_t(i)^{-\theta_t}P_t^{\theta_t}Y_t}{A_t}\right\}$$

yielding the first-order condition:

$$E_{t-1}\beta\frac{C_t^{-\sigma}\xi_t}{P_t}\left\{(1-\theta_t)p_t(i)^{-\theta_t}P_t^{\theta_t}Y_t + \theta_t\frac{W_t P_t^{\theta_t}Y_t}{A_t}p_t(i)^{-\theta_t-1}\right\} = 0$$

Within the two groups, the problem for each firm is the same, i.e. for a fraction γ of firms $p_t(i) = p_t^{flex}$, while $p_t(i) = p_t^{fix}$ for a fraction $1-\gamma$.

As discussed when deriving the Old Keynesian Phillips Curve, to close the supply side you first derive pricing for each firm, and then write the aggregate price index in terms of the component prices. We use the two conditions above to obtain pricing conditions for each firm, obtaining respectively (2) and (3), and then we use the aggregate price index, $P_t \equiv \left[\int_0^1 p_t(i)^{1-\theta}di\right]^{\frac{1}{1-\theta}}$, to get (4):

$$\frac{p_t^{flex}}{P_t} - \frac{\theta}{\theta-1}\frac{W_t}{P_t A_t} = 0 \tag{2}$$

$$E_{t-1}\beta\frac{C_t^{-\sigma}Y_t\xi_t}{P_t^{-\theta}}\left\{\frac{p_t^{fix}}{P_t} - \frac{\theta}{\theta-1}\frac{W_t}{P_t A_t}\right\} = 0 \tag{3}$$

$$P_t^{1-\theta_t} = \gamma(p_t^{flex})^{1-\theta_t} + (1-\gamma)(p_t^{fix})^{1-\theta_t} \tag{4}$$

A first-order approximation of (2), (3) and (4) yields the New Classical Phillips curve in the next proposition:

Proposition 1. ***New Classical Phillips Curve.*** *If a fraction γ of firms set their prices flexibly and $1-\gamma$ one period in advance, then the aggregate supply side of the approximated model is summarized the New Classical Phillips curve:*

$$\pi_t = \kappa(\hat{Y}_t - \hat{Y}_t^n) + E_{t-1}\pi_t \tag{5}$$

where $\kappa \equiv \frac{\gamma}{1-\gamma}\sigma_c^{-1}$.

To prove this proposition we linearize (2), (3) and (4) to obtain:

$$\begin{aligned}
\hat{p}_t^{flex} &= \hat{P}_t + \zeta(\hat{Y}_t - \hat{Y}_t^n) &(6)\\
\hat{p}_t^{fix} &= E_{t-1}\left[\hat{P}_t + \zeta(\hat{Y}_t - \hat{Y}_t^n)\right] &(7)\\
\hat{P}_t &= \gamma\hat{p}_t^{flex} + (1-\gamma)\hat{p}_t^{fix}
\end{aligned}$$

where $\zeta = \sigma_c^{-1}$. The first two conditions imply that $E_{t-1}\hat{p}_t^{flex} = \hat{p}_t^{fix}$. Using the price index: $E_{t-1}\hat{P}_t = \hat{p}_t^{fix}$, so that:

$$\begin{aligned}\hat{P}_t - E_{t-1}\hat{P}_t &= \gamma\hat{p}_t^{flex} + (1-\gamma)\hat{p}_t^{fix} - \hat{p}_t^{fix} \\ &= \gamma(\hat{p}_t^{flex} - \hat{p}_t^{fix})\end{aligned}$$

The price index implies that $\hat{p}_t^{fix} = \frac{1}{1-\gamma}\hat{P}_t - \frac{\gamma}{1-\gamma}\hat{p}_t^{flex}$. Substituting it into the preceding expression, we can write:

$$\pi_t - E_{t-1}\pi_t = \hat{P}_t - E_{t-1}\hat{P}_t = \frac{\gamma}{1-\gamma}(\hat{p}_t^{flex} - \hat{P}_t)$$

Using (6) yields the New Classical Phillips Curve and proves the proposition.

We can also express this Phillips curve in terms of the efficient output gap and the markup shock. If we make the assumption that variations in government spending are efficient and use the relationship between the efficient and natural output from the last chapter, we obtain:

$$\pi_t = \kappa x_t + E_{t-1}\pi_t + \frac{\gamma}{1-\gamma}\hat{\mu}_t \tag{8}$$

This is the **New Classical Phillips Curve**. If inflation is anticipated, it has no effect; if it is unexpected, it increases output. This gives a straightforward interpretation of the breakdown of the Phillips curve in the 1970s that we just documented.

1.2 The New Classical Phillips Curve and the Inflation Bias

We already considered optimal policy with the old-fashioned Phillips curve. What is optimal policy assuming the New Classical Phillips curve? Due to the expectation term, it now becomes critical to be clear about whether we assume that the government can commit to future policy, or that it cannot.

Policy under optimal commitment is when the government can commit fully, from period 0 onward, to its future plans.

A tricky part of formulating this problem is the presence of the lagged expected inflation term in the Phillips curve. This can be solved by defining an auxiliary variable, i.e. the expectation variable $\psi_{t-1} = E_{t-1}\pi_t$ so that $\psi_t = E_t\pi_{t+1}$.

We can now form the Lagrangian for optimal policy under commitment by adding the definition of μ_t as an additional constraint:

$$\mathcal{L}_t = E_0\sum_{t=0}^{\infty}\beta^t\frac{1}{2}\left\{[\pi_t^2 + \lambda_x(x_t - x^*)^2] + \phi_{1t}(\pi_t - \kappa x_t - \frac{\gamma}{1-\gamma}\hat{\mu}_t - \psi_{t-1}) + \phi_{2t}(\psi_t - \pi_{t+1})\right\}$$

The first-order conditions for $t > 0$ (here we do not consider policy at time 0, but we will discuss it later in a more general model) are:

$$\frac{\partial L_t}{\partial \pi_t} = \pi_t + \phi_{1t} - \beta^{-1}\phi_{2t-1} = 0$$
$$\frac{\partial L_t}{\partial \psi_t} = -\beta E_t \phi_{1t+1} + \phi_{2t} = 0$$
$$\frac{\partial L_t}{\partial x_t} = \lambda_x(x_t - x^*) - \kappa\phi_{1t} = 0$$

Making use of the fact that $E_t x_{t+1} = 0$, the first-order conditions can be combined to yield:

$$\pi_t + \frac{\lambda_x}{\kappa} x_t = 0$$

Combining this with equation (5) and assuming the shock is iid yields the solution:

$$x_t = -\frac{\kappa}{\lambda_x + \kappa^2}\frac{\gamma}{1-\gamma}\hat{\mu}_t \tag{9}$$

$$\pi_t = \frac{\lambda_x}{\lambda_x + \kappa^2}\frac{\gamma}{1-\gamma}\hat{\mu}_t \tag{10}$$

Since the shock is iid mean zero, optimal inflation is on average zero. In response to the shock, however, the government will trade off inflation and the output gap. Kydland and Prescott's idea, which won them the Nobel Prize (together with their formulation of the Real Business Cycle Theory), was to characterize the policy of a government that cannot commit to future policy, that is, one that takes any expectations past or current as "given" in setting its policy.

Suppose that the government takes expectations, i.e. ψ_{t-1}, as exogenously given, essentially treating them in the same way as the shock $\hat{\mu}_t$, that is, as completely out of its control.

Then, it solves a period maximization problem:

$$\mathcal{L}_t = E_0 \sum_{t=0}^{\infty} \beta^t \frac{1}{2}\left\{[\pi_t^2 + \lambda_x(x_t - x^*)^2] + \phi_t(\pi_t - \kappa x_t - \frac{\gamma}{1-\gamma}\hat{\mu}_t - \psi_{t-1})\right\}$$

with the FOCs:

$$\frac{\partial L_t}{\partial \pi} = \pi_t + \phi_t = 0$$
$$\frac{\partial L_t}{\partial x_t} = \lambda_x(x_t - x^*) - \phi_t\kappa = 0$$

Combining these yields:

$$\pi_t + \frac{\lambda_x}{\kappa}(x_t - x^*) = 0 \tag{11}$$

We see that the difference relative to the optimal commitment is seen via the presence of x^*. To solve for equilibrium inflation, take the expectation of both sides of equation (5) to obtain that $E_{t-1}x_t = E_{t-1}\mu_t = 0$ by our assumption that μ_t is iid mean zero. Then take the expectation of both sides of condition (11) at time $t-1$ and use $E_{t-1}x_t = 0$ to solve for:

$$E_{t-1}\pi_t = \frac{\lambda_x}{\kappa}x^* \tag{12}$$

Using this to substitute for expectation in the Phillips curve, one can solve for the output gap and inflation using (11) to obtain:

$$x_t = -\frac{\kappa}{\lambda_x + \kappa^2}\frac{\gamma}{1-\gamma}\hat{\mu}_t$$
$$\pi_t = \frac{\lambda_x}{\kappa}x^* + \frac{\lambda_x}{\lambda_x + \kappa^2}\frac{\gamma}{1-\gamma}\hat{\mu}_t$$

Thus there is an increase in inflation without any gain in output relative to output stabilization, which remains the same as under commitment. This provides one explanation for inflation in the 1970s. The government was taking expectations as given, but, with rational expectations, people anticipated this, and the result was higher inflation without any gains in the output gap. At the heart of this explanation of the Great Inflation was that the government could not commit to future policy but was being "discretionary." Kydland and Prescott's solution to this problem was that the Federal Reserve should not set policy under discretion but instead follow "policy rules." Inflation targeting and modern central bank independence are often cited as reactions to Kydland and Prescott's discovery.